

Appendix — Platform Technology Stack



Current



Phase 1



Phase 2



Phase 3+

Cloud Infrastructure

Azure Container Apps	Container hosting, warm pool
AKS	Container orchestration at scale
Azure Service Bus	Async assessment queue
Azure Cache for Redis	Stateless session state
ACR	Immutable artefact registry
Terraform	Infrastructure as Code

Backend & Inference

Python / Flask	Current inference service
FastAPI	Target async inference service
Triton Inference Server	ML model serving at scale
MLflow	Experiment tracking, model registry

Signal Processing

POS + CHROM algorithms	rPPG signal extraction
SNR-weighted fusion	Best signal quality combination
Signal quality gate	Reject low-SNR before transmission

Mobile — iOS

Swift / SwiftUI	iOS SDK language and UI
Apple Vision Framework	Face landmark detection (native)
Metal	GPU compute for signal extraction
Fastlane	Build automation and release

Mobile — Android

Kotlin	Android SDK language
MediaPipe	Cross-platform face landmark tracking
Vulkan	GPU compute for signal extraction

Testing & Verification

pytest	Unit and integration tests
TestRail	Test management, IEC 62304 §5.7 artefact
Renovate / Dependabot	Automated dependency updates
Bland-Altman analysis	Clinical performance validation

Developer Experience

GitHub Pages → Backstage	Docs and internal developer portal
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CI/CD Pipeline

GitHub Actions	Pipeline execution, immutable logs
SonarCloud	Static analysis quality gate
Snyk	Dependency and container vulnerability scan
Syft	SBOM generation (FDA cybersecurity artefact)
Trivy	Container image scanning
AI Code Review Agent	Structured PR review

Observability — Cloud

Grafana	Metrics dashboards and alerting
Prometheus	Infrastructure and service metrics
Loki	Structured log aggregation
Sentry	Error tracking, release correlation
PagerDuty	On-call routing and incident management
Azure Monitor	Native monitoring (considered)

Observability — Mobile

Sentry Mobile SDK	Crash rate, ANR, B2B integration errors
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Appendix — Governance, Security, AI Agents & Regulatory Standards

Governance & QMS

GitHub / GitHub Projects

Source control and sprint planning

Qualio

eQMS — controlled docs, CAPA, training records

ADR format (in repo)

Architecture Decision Records

Slack

Communication only — not a compliance record

Wardley / Value mapping

Value and risk-driven planning

Security

Snyk

Dependency and container vulnerability scanning

Syft

SBOM generation every build

Trivy

Container image scanning

Azure Key Vault

Secrets and certificate management

Cyber Essentials Plus

UK government security baseline

AI Agents

Anthropic Claude API

Primary agentic framework

Code Review Agent

Structured PR review (Z2)

Compliance Agent

Traceability gate at PR time (Z2)

Documentation Agent

Merge-triggered changelog (Z2)

Ops Agent

Alert-triggered issue creation (Z3)

ML Validation Agent

Model drift detection (Z2)

Z2 = Review & CI Gate · Z3 = Operations

Regulatory Standards

IEC 62304

Medical device software lifecycle

ISO 13485

Quality management system

ISO 14971

Risk management

EU MDR Class IIa

CE certification framework

FDA 510(k) / De Novo

US regulatory pathway

GDPR

Data protection

NHS DSPT

NHS data security toolkit

AAMI/ESH/ISO BP standard

Clinical accuracy threshold ($\leq 5 \pm 8$ mmHg)

FDA Cuffless BP Guidance

Jan 2026 clinical performance testing

FDA Cybersecurity Guidance

2023 — SBOM and vulnerability mgmt

Note on SBOM

This SBOM reflects the target platform architecture as described in the presentation. Current state to be validated on day one. In the target architecture, Syft generates a formal machine-readable SBOM (CycloneDX or SPDX format) on every build as a required FDA cybersecurity artefact.